# NEELI PRAVEEN

praveen_neeli@srmap.edu.in | 9392801318 | 2027
M.Tech, Data Science
Address: 1-35, back side ysr bomma
LinkedIn | GitHub

## Skills Summary

**Languages:** Python, SQL, Java, HTML/CSS
**Big Data & Cloud:** Apache Spark, Kafka, Hadoop (HDFS, YARN, MapReduce)
**Machine Learning:** Deep Learning (Transformers,CNNs), NLP, Scikit-learn, TensorFlow/PyTorch, SHAP
**Libraries & Tools:** Pandas, NumPy, Matplotlib, Git, Jupyter Notebook, VS Code, Web Scraping

## Education

**SRM, Amaravati** — July 2025 – Ongoing
- Masters in Data Science — **SGPA - 9.43**

**Natioanal Institute of Technology, Andhra Pradesh** — May 2024
- Bachelors in Computer Science and Engneering — **First Class**

**Sri Chaitanya Junior college, Vijayawada** — March 2020
- Andhra Pradesh Board of Intermediate Education — **CGPA - 9.75**

**Sri Chaitanya High School, Mogalrajpuram** — March 2018
- Andhra Pradesh Board of Secondary Education — **CGPA - 9.8**

## Projects

### Evaluating CNNs and Transformers for Multi-Label Classification in 3D CT Imaging — Jul 2024

- Engineered a **segmentation-guided deep learning framework** to detect cervical spine fractures in 3D CT scans, evaluating 8 architectures including **MViT and Swin Transformers** for optimal diagnostic precision.
- Addressed extreme class imbalance within the **RSNA 2022 medical dataset** by implementing **Weighted Binary Cross-Entropy and Focal Loss** to enhance model reliability and detection sensitivity.

### Stabilizer V2: Hybrid Deep Learning for High-Confidence Heart Disease Prediction — Sep 2025

- Developed the **Stabilizer V2 Protocol** by merging **Transformers with Gated Residual Networks (GRN)** and leveraging **TVAE-based data augmentation** to expand the training set to 5,000 records.
- Integrated **Monte Carlo Dropout** for predictive uncertainty quantification and utilized **SHAP** to ensure transparency and alignment with clinical indicators like chest pain type.

### Distributed Anomaly Detection System for Time-Series Data using Big Data Pipeline — Dec 2025

- Developed a **high-performance pipeline** using Apache Flume, HDFS, and PySpark to process 67 concurrent data streams in parallelArchitected a **novel Attention-augmented Deep Learning model** (CNN + BiLSTM + Self-Attention) to detect anomalies in noisy industrial sensor data
- Implemented a **leakage-free dynamic thresholding strategy** using Exponential Weighted Moving Averages (EWMA) to improve detection robustnessAchieved good performance status ($F1 \geq 0.6$) on nearly **50 pencentage of tested datasets**, successfully automating model retraining and HDFS artifact storage

## Certifications

### NPTEL – Applied Accelerated Artificial Intelligence — October 2023

- Studied **NPTEL certification course on "Applied Accelerated Artificial Intelligence," with silver grade organized by IIT Madras**

### Published Researcher: RAIT 2025 Conference — May 2025

- Published "**Evaluating CNNs and Transformers for Multi-Label Classification in 3D CT Imaging**"at the 6th International Conference on Recent Advances in Information Technology (RAIT)